Graphene conductivity

The goal is to establish the residual conductivity of graphene, within linear response, in a non-interacting model.

Setting. Spinless electrons, atomic units. Let $\mathcal{R} \subset \mathbb{R}^2$ be a lattice, Ω a unit cell, and $\mathcal{B}$ a unit cell of

$$\mathcal{R}^* = \{K \in \mathbb{R}^2 : K \cdot R \in 2\pi\mathbb{Z} \text{ for every } R \in \mathcal{R}\}.$$

For real $V \in H^1_{\mathrm{per}}(\Omega)$, set

$$H = -\tfrac{1}{2}\Delta + V \quad \text{on } L^2(\mathbb{R}^2;\mathbb{C}), \qquad H_k = \tfrac{1}{2}(-i\nabla + k)^2 + V \quad \text{on } L^2_{\mathrm{per}}(\Omega;\mathbb{C}).$$

H_k is self-adjoint with domain $H^2_{\mathrm{per}}(\Omega;\mathbb{C})$. Write its eigenvalues, with multiplicities, as $\lambda_1(k) \le \lambda_2(k) \le \cdots$, and set $\lambda_0(k) = -\infty$.

Band assumptions. Fix $\mu_{\mathrm{F}} \in \mathbb{R}$, and assume that the system is semi-metallic: there are distinct points $k_1, \dots, k_M$ modulo $\mathcal{R}^*$ and a number of bands N such that

$$\lambda_N(k) < \mu_{\mathrm{F}} < \lambda_{N+1}(k) \qquad (k \notin \{k_1, \dots, k_M\} \bmod \mathcal{R}^*),$$

and, for each $i = 1, \dots, M$,

$$\lambda_{N-1}(k_i) < \lambda_N(k_i) = \mu_{\mathrm{F}} = \lambda_{N+1}(k_i) < \lambda_{N+2}(k_i).$$

There are $v_i > 0$ such that, as $q \to 0$,

$$\lambda_N(k_i + q) = \mu_{\mathrm{F}} - v_i|q| + O(|q|^2), \qquad \lambda_{N+1}(k_i + q) = \mu_{\mathrm{F}} + v_i|q| + O(|q|^2).$$

Current. Fix $e_\beta, e_\alpha \in \mathbb{R}^2$, with e_β the field direction and e_α the current direction. For all $\varepsilon \ge 0$ and $t \ge 0$, define

$$\gamma_0 = \mathbf{1}_{(-\infty,\mu_{\mathrm{F}}]}(H), \qquad H_\varepsilon = H - \varepsilon x \cdot e_\beta, \qquad J_\alpha = (-i\nabla) \cdot e_\alpha,$$
$$\gamma_\varepsilon(t) = e^{-itH_\varepsilon}\gamma_0 e^{itH_\varepsilon}, \qquad j_\varepsilon(t) = \underline{\mathrm{Tr}}\big(J_\alpha \gamma_\varepsilon(t)\big),$$

where H_ε denotes the self-adjoint closure of its restriction to $C^\infty_c(\mathbb{R}^2;\mathbb{C})$, and

$$\underline{\mathrm{Tr}}(A) = \frac{1}{|\Omega|}\,\mathrm{Tr}(\mathbf{1}_\Omega A \mathbf{1}_\Omega)$$

for locally trace-class periodic A.

Theorem. *Under the above assumptions, the iterated limit exists and*

$$\lim_{T\to+\infty} \frac{1}{T}\int_0^T \left(\lim_{\varepsilon\to 0^+} \frac{j_\varepsilon(t)}{\varepsilon}\right) dt = \frac{M}{16}\, e_\alpha \cdot e_\beta.$$